IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

POWER ELECTRONICS AND POWER SYSTEMS

Friday, 10 May 2:30 pm

Time allowed: 2:00 hours

There are FIVE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : Y. Gu, M. O'Malley

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Information for Candidates

Switch-Mode Power Supplies

Voltage ratio equation for buck SMPS in continuous mode $\frac{V_O}{V_I} = \delta$

Voltage ratio equations for boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{1}{1-\delta}$

Voltage ratio equations for buck-boost SMPS in continuous mode $\frac{V_O}{V_I} = \frac{-\delta}{1-\delta}$

Three-Phase Systems (Balanced)

Line/Phase Voltages and Currents

Star $V_L = \sqrt{3}\, V_P;\ I_L = I_P$ Delta $V_L = V_P;\ I_L = \sqrt{3}\, I_P$

Power $P_{3\phi} = 3\, V_P I_P cos(\phi) = \sqrt{3}\ V_L I_L cos(\phi)$

Transformers

Ideal transformer $\bar{V}_2 = k_N \bar{V}_1$; $\bar{I}_2{}' = k_N \bar{I}_2$

Impedances referral $\bar{Z}'_2 = {\bar{Z}_2}/{k_N{}^2}$

Voltage regulation $\Delta V_2 = \frac{\left|V_2^{o/c}\right| - |V_2^{load}|}{\left|V_2^{o/c}\right|}$

Induction Machines

$\omega_S = {\omega_E}/{P}$; $\quad s = \frac{\omega_S - \omega_R}{\omega_S}$; $\quad T_{EM} = \frac{3\, I'^2_R\, R'_R}{\omega_R}\left(\frac{1-s}{s}\right)$

Power Flow in Lines and Cables

Cable Parameters (expressed per metre)

$R'_{LF} = \frac{1}{\sigma_C \pi\, r_C^2} + \frac{1}{\sigma_C 2\pi\, r_O\, t_O}$ $\quad L' = \frac{\mu}{2\pi} ln\left({r_O}/{r_C}\right)$ $\quad C' = \frac{2\pi\varepsilon_0\varepsilon_{RI}}{ln\left({r_O}/{r_C}\right)}$ $\quad G' = \frac{2\pi\sigma_I}{ln\left({r_O}/{r_C}\right)}$

OHL Parameters in approximate form (expressed per metre)

$R'_{LF} = \frac{2}{\sigma_C \pi\, r_C^2}$ $\quad L' = \frac{\mu}{\pi} ln\left({d}/{r_C}\right)$ $\quad C' = \frac{\pi\varepsilon_0\varepsilon_{RI}}{ln\left({d}/{r_C}\right)}$ $\quad G' = \frac{\pi\sigma_I}{ln\left({d}/{r_C}\right)}$

Power Flow (full form)

$$P_S = \frac{{V_S}^2}{Z_{SR}}\cos(\theta) - \frac{V_R V_S}{Z_{SR}}\cos(\theta+\delta) \qquad Q_S = \frac{{V_S}^2}{Z_{SR}}\sin(\theta) - \frac{V_R V_S}{Z_{SR}}\sin(\theta+\delta)$$

Power Flow assuming inductance-dominated impedance

$$P_S \simeq \frac{V_R V_S}{X_{SR}}\sin(\delta) \qquad Q_S \simeq \frac{V_S}{X_{SR}}\left(V_S - V_R\cos(\delta)\right)$$

Voltage Drop (approximate form) $\Delta V = |V_S| - |V_R| \approx \frac{R\, P_S + X\, Q_S}{|V_S|}$

1. This question addresses three-phase power systems.

   As shown in Figure 1, a $100\sqrt{2}$ V, three-phase 50Hz is connected across a balanced delta connected load of three impedances each consisting of a 10 Ω resistor in series with a capacitor of $(1/1000\pi)$ F .

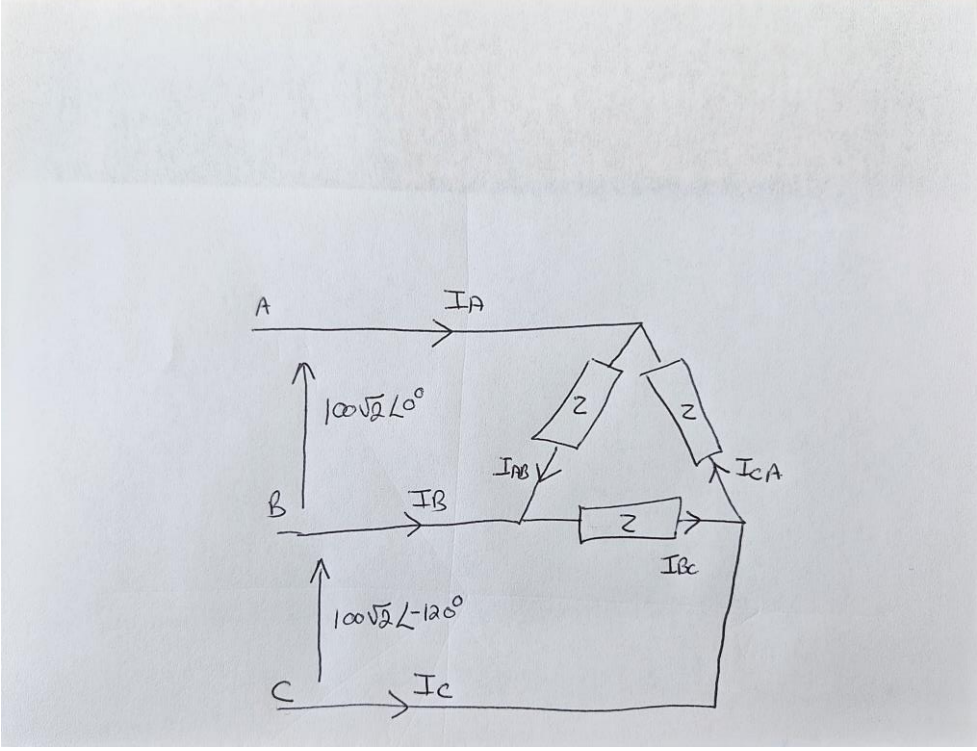


Figure 1

   (i) Calculate the phase currents [2]

   (ii) Calculate the line currents [2]

   (iii) Sketch the line and phase currents and voltages on a phasor diagram [4]

   (iv) Calculate the total (all three phases) real and reactive power drawn [2]

2. This question addresses transformers.

   (a) Explain eddy currents and how they affect the operation of a transformer and how they can be reduced. [4]

   (b) A transformer nameplate indicates that it is intended for use as 6 kV to 500 V transformer at a rated apparent power of 180 kVA. The approximate equivalent circuit impedance properties of the transformer were measured as follows.

Combined winding resistance referred to primary	R_W = 3 Ω
Combined leakage reactance referred to primary	X_W = 4Ω
Magnetising resistance	X_M = 10 kΩ
Magnetising reactance	R_M = 1 kΩ

(i) Estimate the turns ratio from the quoted operating voltages i.e. 6 kV and 500 V. [2]

(ii) Calculate the rated current on both the primary and secondary side. [2]

(iii) Draw the approximate equivalent circuit of the transformer [3]

(iv) Calculate the secondary voltage when a rated (6kV) voltage is applied to the primary side and the secondary terminals are open circuit [2]

(v) Calculate the secondary current magnitude when a 150 V voltage is applied to the primary side and the secondary terminals are short circuit. [2]

(vi) Why would increasing the voltage above 150 V in these circumstances (secondary side short circuit) not be recommended. [2]

(c) A load of 2 Ω is put on the secondary side and the rated voltage (6kV) is applied on the primary side. Calculate the

(i) magnitude of the current drawn by this load [3]

(ii) the voltage drop across the load [2]

(iii) the power dissipated in the load [2]

(iv) losses in the transformer [4]

(v) efficiency of the transformer [2]

3. This question addresses induction machines.

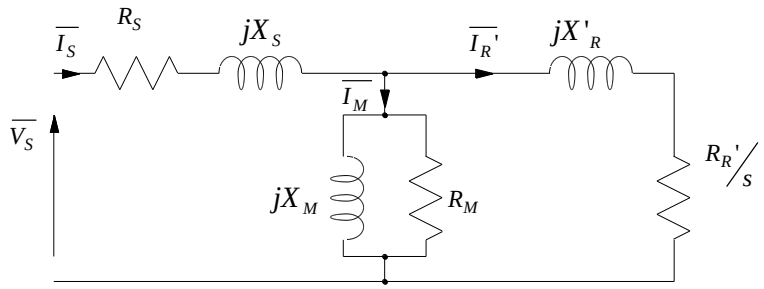


Figure 2

As shown in Figure 2, a 4-pole, 3-phase induction machine is connected in star to a 480 V (line), 50 Hz supply. The equivalent circuit parameters, referred to the stator, are:

Stator resistance	0.05 Ω
Referred rotor resistance	0.06 Ω
Stator leakage reactance	2.0 Ω
Referred rotor leakage reactance	1.0 Ω
Magnetising reactance	40.0 Ω
Magnetising resistance	200 Ω

For operation at 1,450 r.p.m. calculate the following:

(i) the slip; [2]

(ii) the stator current; [8]

(iii) the power factor stating whether it is lagging or leading and any observation on its value with respect to what might be desirable i.e. power factor close to unity [2]

(iv) ignoring the magnetizing reactance and resistance calculate the electromagnetic torque [4]

(v) If the electromagnetic torque halves, without performing detailed calculations but noting any assumptions estimate the mechanical speed [4]

4. This question addresses power flow in lines and cables.

(a) Consider an overhead line used for a single-phase AC supply. The line has a series impedance of $j20\ \Omega$, that is, it is purely inductive. Calculate the real power at the sending end if the sending-end and receiving-end voltages are both 220 kV and the receiving-end voltage is lagging the sending-end voltage by 10.0°. [4]

(b) The line has a maximum current carrying capability of 4,000 A. For operation of this line under this limit, and with both ends at 220 kV, calculate the load angle at which the line would operate and the power transfer this creates. [3]

(c) An overhead line has a series impedance of 0.5 + j 0.4 Ω. The voltage at the sending end is 11 kV. A real power of +8.0 MW is measured flowing into the line at the sending end. The voltage at the receiving end is measured at 11kV. Estimate the reactive power as seen at the sending end. You may use approximations in your calculation. [3]

5. This question addresses switching mode power supplies.

A **buck** SMPS is to be designed to convert 20.0 V to 5.0 V with maximum output currents of 10.0 A. The on-state resistance of the chosen MOSFET is 0.02Ω. The diode is an ideal diode with no forward volage and no on-state resistance.

(a) Calculate duty cycle needed for the 20.0 V to 5.0 V voltage conversion ratio, neglecting the voltage drop on the MOSFET and diode. [5]

(b) The converter is in critical conduction mode at the output current of 1A. The switching frequency is 10kHz. Calculate the inductance. [5]

(c) Based on (b), calculate the ripple current on the inductor at continuous conduction mode. [5]

(d) Based on (c), calculate the ripple voltage on the equivalent series resistor (ESR) of the capacitor if the ESR=10mΩ. [5]

(e) The turn-on and turn-off energy losses of the MOSFET are 25.0 μJ and 25.0 μJ respectively. The switching frequency is 10kHz. Calculate the switching loss on the MOSFET. [3]

(f) Calculate the conduction loss on the MOSFET at the maximum output current (10.0A), assuming the converter is in continuous conduction mode and the ripple current is negligible. [3]

(g) The MOSFET is mounted on a heat-sink which gives a total thermal impedance between junction and ambient of $R_{ThJA} = 30\ °C/W$. The ambient temperature is 40°C. Suppose the total power loss on the MOSFET is 1W, calculate the junction temperature. [4]